Problem K. Sequence Shift

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2.5 seconds
Memory limit:	512 megabytes

You are given two sequences of length n: $[a_1, a_2, \ldots, a_n]$ and $[b_1, b_2, \ldots, b_n]$. The value of $f(a, b)$ is defined as $f(a, b) = \max\{a_i + b_i\}$, where $1 \le i \le n$.

The sequence b can be shifted. You will then be given q operations, each operation can be divided into the following two steps:

- First, shift the sequence b to the left by one position, and drop the first element, so the sequence b' will be $[b'_1 = b_2, b'_2 = b_3, \ldots, b'_{n-1} = b_n]$.
- Then, append v to the rightmost place of b, so the sequence b' will be $[b'_1 = b_2, b'_2 = b_3, \ldots, b'_{n-1} = b_n, b'_n = v]$.

In this problem, your task is to figure out the value of $f(a, b)$ before/after each operation.

Input

The first line of the input contains two integers n and q ($1 \le n \le 1\,000\,000$, $0 \le q \le 1\,000\,000$), denoting the length of the sequences and the number of operations.

The second line contains n integers $a_1, a_2, \ldots, a_n$, denoting the sequence a.

The third line contains n integers $b_1, b_2, \ldots, b_n$, denoting the initial sequence b.

Each of the next q lines contains a single integer v, denoting the value that will be appended in each operation. The value of v will be encrypted in order to enforce online processing.

It is guaranteed that all the values of a_i, b_i and v are chosen uniformly at random from integers in the range $[1, 10^9]$. The randomness condition does not apply to the sample test(s), but your solution must pass the sample test(s) as well.

Let *last* be the previous value of $f(a, b)$ that you answered. For each operation, the actual value of v is $v \oplus last$. In the expressions above, the symbol "$\oplus$" denotes the bitwise exclusive-or operation. Also, note that the constraints described in the statement above apply to the corresponding parameters only after decryption, the encrypted values are not subject to those constraints.

Output

Print $q + 1$ lines.

Output a single integer in the first line, denoting the initial value of $f(a, b)$.

In the k-th line ($2 \le k \le q + 1$), output a single integer denoting the current value of $f(a, b)$ after the $(k - 1)$-th operation.

Example

standard input	standard output
5 3 1 4 3 2 5 7 5 8 3 2 3 6 4	11 13 16 25